

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0518

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

(b) Apply a negative transformation $s = 255 - r$ to the pixel values: $r = [50, 100, 150, 200]$

Compute the transformed values.

Grey Level Transformation

(a) Purpose

Grey level transformation is used to improve image quality by changing pixel intensities. It enhances contrast, highlights details, and makes objects more visible.

(b) Negative transformation

Formula:

$$s = L - 1 - r$$

where:

s = output pixel value

r = input pixel value

L = number of intensity levels

For an 8-bit image ($L=256$):

$$s = 255 - r$$

Example:

$$r=50 \rightarrow s=205$$

$$r=100 \rightarrow s=155$$

$$r=200 \rightarrow s=55$$

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

(b) Given $s = c \log(1 + r)$, where $c = 10$, compute s for:

$$r = [0, 10, 50, 100]$$

2. Log Transformation

(a) Purpose

Log transformation expands low-intensity values and compresses high-intensity values, making more visible.

(b) Formula

$$s = c \log(1 + r)$$

Example ($c = 1$):

- $r = 10$

$$s = \log(11) = 2.39$$

- $r = 50$

$$s = \log(51) = 3.93$$

- $r = 100$

$$s = \log(101) = 4.61$$

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

(b) Given a 3-bit image with histogram:

Intensity	Frequency
0	2
1	2
2	4
3	8

Compute the cumulative distribution and equalized intensity values.

3. Histogram Processing

(a) Purpose

Histogram equalization improves image contrast by redistributing intensity values.

(b) Given histogram

Intensity	Frequency
0	2
1	2
2	4
3	8

Total pixels:

$$N=2+2+4+8=16$$

Cumulative distribution:

Intensity	CDF
0	2
1	4
2	8
3	16

Equalized values:

$$sk=16/CDF \times 3$$

Intensity	Equalized value
0	0
1	1
2	2
3	3

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

(b) Apply a 3×3 averaging filter to the following region:

[10 20 30 20 30 40 30 40 50]

Compute the new center pixel value.

4. Spatial Filtering (Smoothing)

(a) Purpose

Smoothing removes noise and small variations from an image.

(b) Averaging filter

For a 3×3 mask:

$$\text{Output} = \frac{\text{sum of all pixels}}{9}$$

Example:

$$\begin{bmatrix} 10 & 20 & 30 \\ 40 & 50 & 60 \\ 70 & 80 & 90 \end{bmatrix}$$

Sum:

$$10 + 20 + 30 + 40 + 50 + 60 + 70 + 80 + 90 = 450$$

Output:

$$450/9 = 50$$

New center pixel = 50



5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

to the matrix:

$$\begin{bmatrix} 10 & 10 & 10 & 10 & 20 & 10 & 10 & 10 & 10 \end{bmatrix}$$

Compute the output at the center.

5. Spatial Filtering (Sharpening)

(a) Purpose

Sharpening enhances edges and fine details.

(b) Laplacian mask

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

For

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 50 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

Output:

$$(4 \times 50) - (10 + 10 + 10 + 10)$$

$$200 - 40 = 160$$

Center output = 160



6. Frequency Domain Filtering

(a) Differentiate low-pass and high-pass filtering in frequency domain.

(b) Given frequency components:

$$F = [100, 80, 40, 20]$$

Apply a low-pass filter that retains only the first two components. Write the filtered output.

6. Frequency Domain Filtering

(a) Difference

Low-pass filter	High-pass filter
Removes noise	Enhances edges
Retains low frequencies	Retains high frequencies
Smooths image	Sharpens image

(b) Example

Frequency components:

[20, 15, 10, 5]

After applying a low-pass filter:

[20, 15, 0, 0]

7. Thresholding

(a) Explain global thresholding in segmentation.

(b) Given pixel values:

[200]

Apply threshold $T = 100$. Assign values (0 or 1).

7. Thresholding

(a) Global thresholding

Global thresholding divides an image into foreground and background using one threshold value.

(b) Example

Pixels:

[20, 60, 80, 30]

Threshold:

$T = 50$

Output:

[0, 1, 1, 0]

8. Edge Detection

(a) Explain the concept of edge detection.

(b) Apply the Sobel operator (horizontal):

$$[-1 \ -2 \ -1, \quad 0 \ 0 \ 0, \ 1 \ 2 \ 1]$$

to:

$$[10 \ 10 \ 10, \ 20 \ 20 \ 20, \ 30 \ 30 \ 30]$$

Compute the gradient at the center.

(a) Purpose

Edge detection identifies boundaries where intensity changes rapidly.

(b) Sobel operator

Horizontal mask:

$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Example:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Gradient:

$$\begin{aligned} &(-1)(1) + 0(2) + (1)(3) \\ &+ (-2)(4) + 0(5) + (2)(6) \\ &+ (-1)(7) + 0(8) + (1)(9) \end{aligned}$$

$$2 + 4 + 2 = 8$$

Gradient = 8



9. Region-Based Segmentation

(a) Explain region growing in segmentation.

(b) Given pixel intensities: $[1 \ 0 \ 2]$

If seed = 50 and threshold difference = 5, identify pixels belonging to the region.

9. Region-Based Segmentation

(a) Purpose

Region growing starts from a seed pixel and adds neighboring pixels with similar intensities.

(b) Example

Seed = 50

Threshold difference = 5

Region includes:

45, 47, 49, 50, 52, 54, 55

These pixels belong to the same region.

10. Combined Enhancement & Segmentation

(a) Explain why smoothing is applied before segmentation.

(b) Given image values:

[1 0 5]

First apply averaging (assume mean = 48), then threshold at $T = 50$.

Determine final segmented output.

Combined Enhancement and Segmentation

(a) Purpose

Smoothing removes noise before segmentation and improves accuracy.

(b) Process

Mean value after averaging:

48

Threshold:

$T=50$

Segmentation rule:

- Pixel $< 50 \rightarrow 0$
- Pixel $\geq 50 \rightarrow 1$

Example:

[40,48,55,60]

Output:

[0,0,1,1]

Thus, the final segmented image is: [0,0,1,1]

Code of Conduct:

I P.V.N. ANISH BABU (192472329) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.V.N. Anish